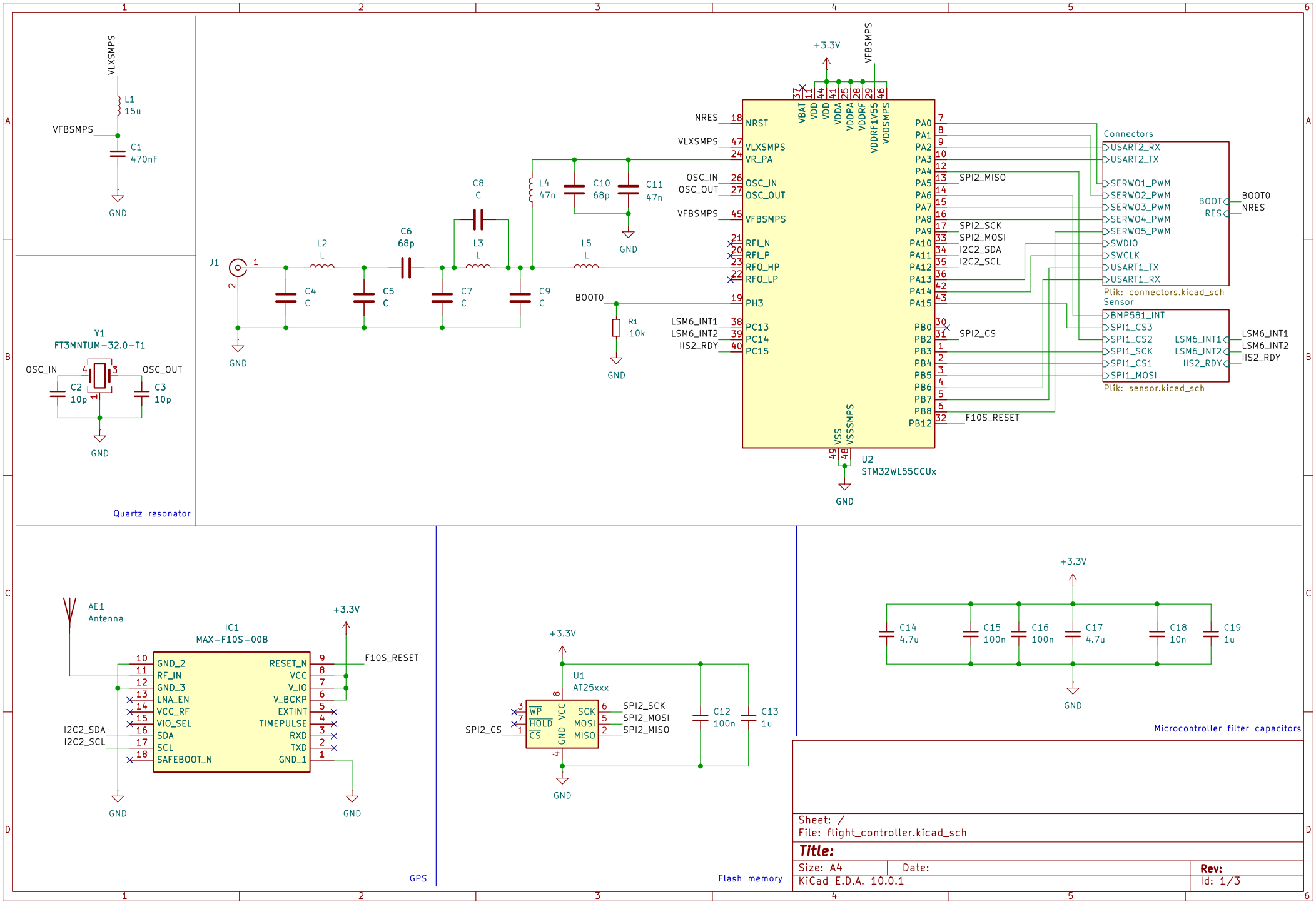
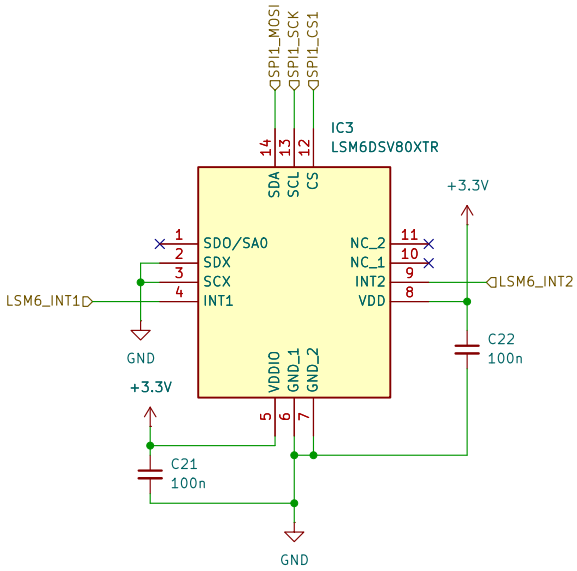
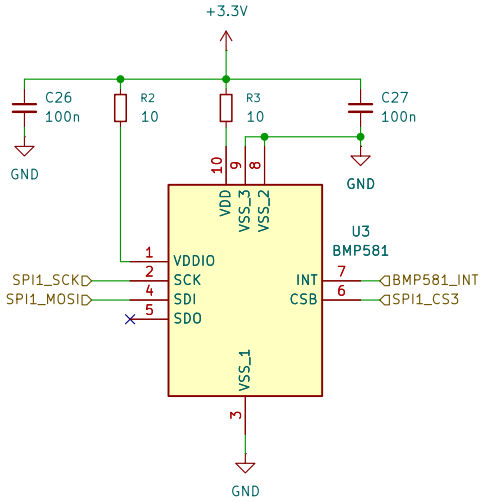
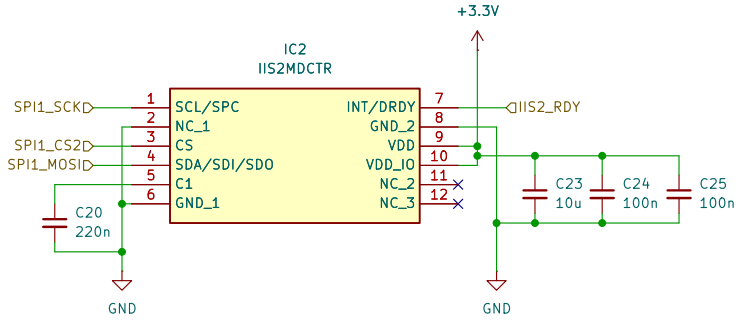

VLXSMPS
L1
15u
VFBSMPS
C1
470nF
GND
Y1
FT3MNTUM-32.0-T1
OSC_IN
OSC_OUT
C2
10p
C3
10p
Quartz resonator
J1
L2
C6
68p
L3
C8
L4
47n
C10
68p
C11
47n
L5
C4
C5
C7
C9
BOOT0
R1
10k
+3.3V
VFBSMPS
NRES
VLXSMPS
OSC_IN
OSC_OUT
VFBSMPS
RFI_N
RFI_P
RFO_HP
RFO_LP
PH3
LSM6_INT1
LSM6_INT2
IIS2_RDY
PC13
PC14
PC15
VBAT
VDD
VDDA
VDDPA
VDDRF
VDDRF1V55
VDDSMPS
VSS
VSSSMPS
U2
STM32WL55CCUx
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB12
SPI2_MISO
SPI2_SCK
SPI2_MOSI
I2C2_SDA
I2C2_SCL
SPI2_CS
F10S_RESET
Connectors
USART2_RX
USART2_TX
SERWO1_PWM
SERWO2_PWM
SERWO3_PWM
SERWO4_PWM
SERWO5_PWM
SWDIO
SWCLK
USART1_TX
USART1_RX
BOOT
RES
BOOT0
NRES
Plik: connectors.kicad_sch
Sensor
BMP581_INT
SPI1_CS3
SPI1_CS2
SPI1_SCK
SPI1_CS1
SPI1_MOSI
LSM6_INT1
LSM6_INT2
IIS2_RDY
Plik: sensor.kicad_sch
AE1
Antenna
IC1
MAX-F10S-00B
GND_2
RF_IN
GND_3
LNA_EN
VCC_RF
VIO_SEL
SDA
SCL
SAFEBOOT_N
RESET_N
VCC
V_IO
V_BCKP
EXTINT
TIMEPULSE
RXD
TXD
GND_1
GPS
U1
AT25xxx
WP
HOLD
CS
SCK
MOSI
MISO
C12
100n
C13
1u
Flash memory
C14
4.7u
C15
100n
C16
100n
C17
4.7u
C18
10n
C19
1u
Microcontroller filter capacitors
Sheet: /
File: flight_controller.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.1
Id: 1/3



Accelerometer, gyroscope



Barometr

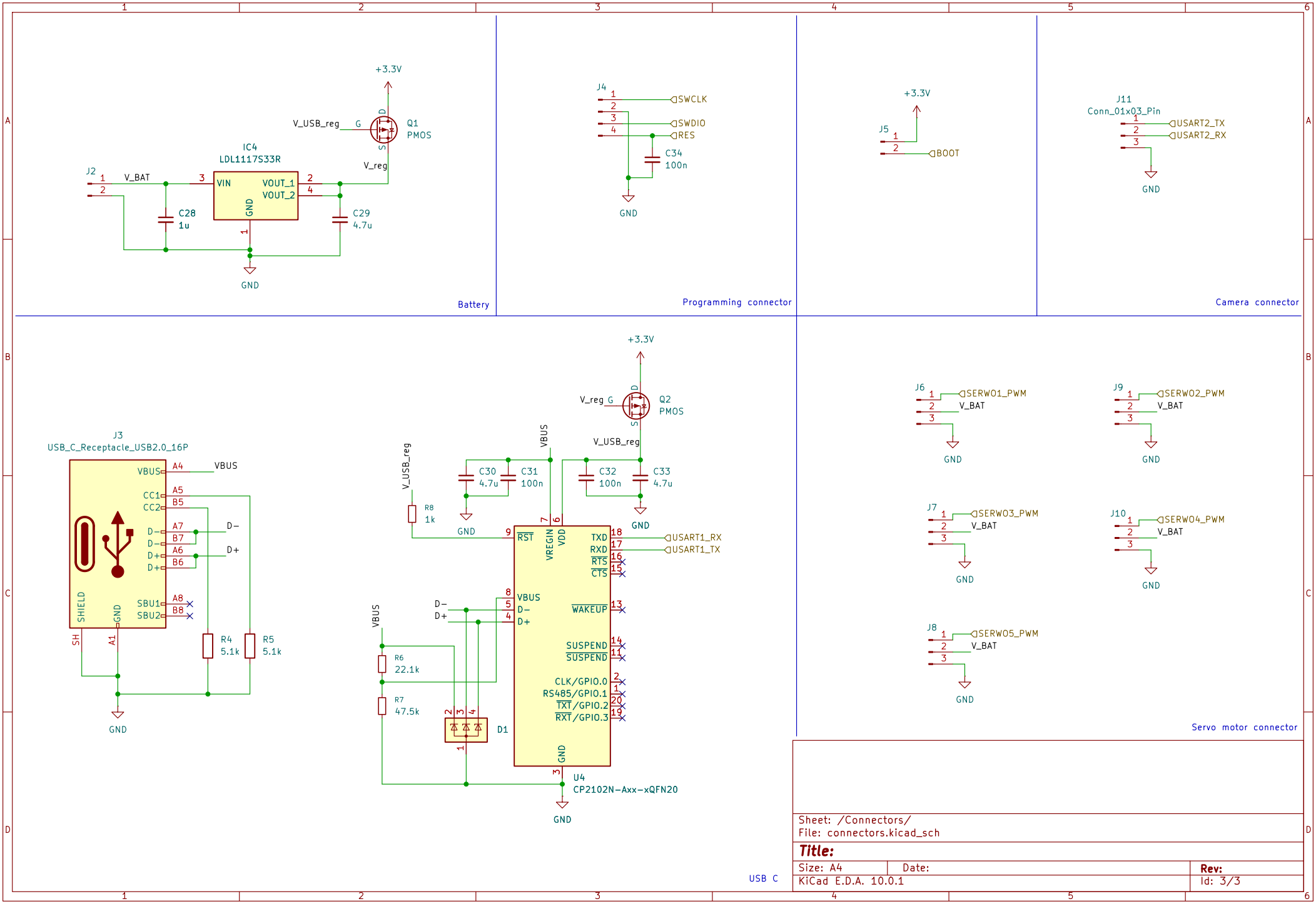


Magnetometer

Sheet: /Sensor/
File: sensor.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.1 | Id: 2/3


Battery
+3.3V
V_USB_reg
G
Q1
PMOS
D
S
IC4
LDL1117S33R
J2
V_BAT
VIN
VOUT_1
VOUT_2
GND
V_reg
C28
1u
C29
4.7u
GND
Programming connector
J4
SWCLK
SWDIO
RES
C34
100n
GND
+3.3V
J5
BOOT
Camera connector
J11
Conn_01x03_Pin
USART2_TX
USART2_RX
GND
J3
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D−
D+
SHIELD
GND
SBU1
SBU2
SH
A1
R4
5.1k
R5
5.1k
GND
+3.3V
V_reg
G
Q2
PMOS
V_USB_reg
VBUS
C30
4.7u
C31
100n
C32
100n
C33
4.7u
R8
1k
GND
RST
VREGIN
VDD
TXD
RXD
RTS
CTS
USART1_RX
USART1_TX
WAKEUP
SUSPEND
SUSPEND
CLK/GPIO.0
RS485/GPIO.1
TXT/GPIO.2
RXT/GPIO.3
R6
22.1k
R7
47.5k
D1
U4
CP2102N−Axx−xQFN20
GND
Servo motor connector
J6
SERWO1_PWM
V_BAT
J9
SERWO2_PWM
J7
SERWO3_PWM
J10
SERWO4_PWM
J8
SERWO5_PWM
GND
USB C
Sheet: /Connectors/
File: connectors.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.1
Id: 3/3